

ASSESSMENT – 1

CompTIA A+

Hardware Components &
System Assembly

Section A — Concept Application

SCENARIO 1: You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

➤ To select compatible hardware for an office workstation:

1. **Motherboard:** Choose one with the correct CPU socket, RAM type, and storage interfaces.
2. **CPU:** Make sure the CPU socket and chipset are supported by the motherboard.
3. **RAM:** Check DDR type (DDR4/DDR5), speed, capacity, and available slots.
4. **Storage:** Check whether the motherboard supports **M.2 NVMe** or **SATA**.
5. **Confirm Compatibility:** Check the manufacturer's specifications, CPU support list, RAM compatibility, available slots, and power requirements.

- **Conclusion:** All components must have matching sockets, interfaces, standards, and power requirements to work together.

SCENARIO 2: You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

➤ **HDD vs SSD vs NVMe**

Storage	Technology	Speed	Cost	Best Use
HDD	Magnetic disks	Slow	Low	Large, low-cost storage
SSD	Flash memory, SATA	Fast	Medium	Office and general use
NVMe SSD	Flash memory, PCIe	Very fast	Higher	High-performance workloads

➤ **Recommendation:**

For office employees using **productivity applications**, I recommend an **NVMe SSD**.

➤ **Technical reasons:**

- Much faster read/write speeds than HDD and SATA SSD.
- Low latency, so applications and files open quickly.

- Uses the **PCIe interface**, providing higher bandwidth.
- Improves boot time and multitasking performance.
- No moving parts, making it quieter and more reliable than HDD.

➤ **Conclusion:** An **NVMe SSD (e.g., 500 GB–1 TB)** provides the best overall performance for modern office workstations.

SCENARIO 3: You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

➤ **Explanation:**

- **GPU (Graphics Processing Unit):** Handles graphics and visual calculations for design software. A suitable GPU provides smooth performance, faster rendering, and better handling of 3D/graphics tasks.
- **Cooling System:** Removes heat produced by the GPU and CPU during long hours of use. Good cooling prevents overheating, thermal throttling, crashes, and hardware damage.

➤ **Conclusion:** A capable GPU plus effective cooling ensures stable, smooth performance during sustained workloads.

SCENARIO 4: You are assembling a PC from components sourced from three different vendors.

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

➤ **Hardware Compatibility in PC Assembly:**

Hardware compatibility is important because all components must work together correctly to make the computer function properly.

➤ **Why it matters:**

- Ensures the **CPU fits and is supported by the motherboard.**
- Ensures the **RAM type and speed** are supported.
- Ensures the **power supply (PSU)** provides enough power.
- Ensures the **storage devices** are compatible with available SATA or M.2 slots.
- Ensures the **GPU fits** the motherboard and PC case.
- Prevents performance and stability problems.

➤ **Problems caused by incompatible components:**

1. PC may **not start or boot**.
2. RAM may **not be detected**.
3. CPU may **not work** with the motherboard.
4. Components may **overheat**.
5. PSU may be **insufficient**, causing crashes or shutdowns.
6. Hardware may **not physically fit** inside the case.
7. Some features may **not work properly** due to incompatible drivers or interfaces.

➤ **Conclusion:** Before buying components from different vendors, check their **socket, supported standards, dimensions, power requirements, and motherboard compatibility**.

SCENARIO 5: You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

➤ **PSU Selection and Safety Checks:**

1. Choose the Right PSU

- Calculate the total power required by the CPU, GPU, motherboard, and storage.
- Choose a PSU with extra power *around 20–30% headroom*.
- Check that the PSU has the required connectors.
- Select a reliable PSU with safety protections.

2. Safety Checks

- Switch off and unplug the PSU before making connections.
- Check cables and components for damage.
- Use anti-static precautions.
- Use only cables compatible with the PSU.

3. Connect the PSU

- Connect the **24-pin cable** to the motherboard.
- Connect the **8-pin CPU cable** to the motherboard.
- Connect the required power cables to the GPU.
- Connect SATA power cables to SSDs and HDDs.
- Ensure all connections are tight and secure.

4. Before First Boot

- Check that the motherboard and components are installed correctly.
- Ensure there are no loose screws or exposed wires.
- Check that cables are away from fans.
- Confirm all power connectors are properly inserted.
- Connect the AC power cable and switch on the PSU.

➤ **Conclusion:** Choose a PSU with sufficient wattage and compatible connectors, then check all connections and safety precautions before powering on the computer.

SCENARIO 6: You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

➤ **POST Checks:**

1. **Switch on the computer** and check that fans and lights work.
2. Check that the **display shows the BIOS/POST screen**.
3. Make sure there are **no error messages or beep codes**.
4. Enter BIOS and check that **RAM, CPU, storage, and GPU** are detected.
5. Confirm the computer **boots into the operating system normally**.

➤ **If POST Fails:**

- Switch off and unplug the computer.
- Check all **power and component connections**.
- Reseat the **RAM and GPU**.
- Check **beep codes or motherboard debug lights**.

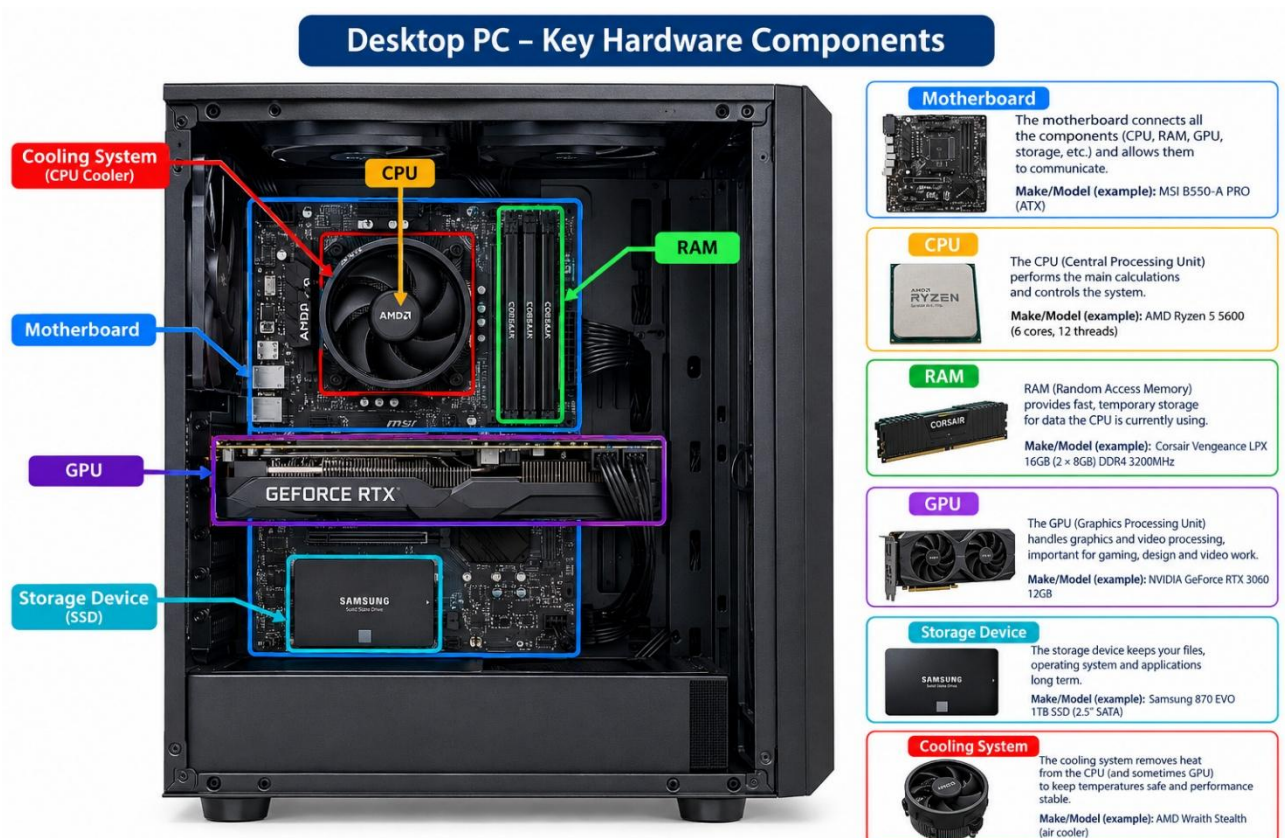
- Reset the **BIOS/CMOS** if needed.
- Test or replace the suspected faulty component.

➤ **Conclusion:** The workstation is ready for the client when it **passes POST, detects all major hardware, and boots into the operating system without errors.**

Section B — Practical Coding Tasks

Task 1: Component Identification & Functions

- Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.
- Note the make/model or key specification of each component where visible.
- Present the findings as a labeled component list or table.



➤ Desktop PC Components:

No.	Component	Function	Make/Model or Key Specification
1	Motherboard	Connects all PC components and allows communication.	MSI B450 TOMAHAWK
2	CPU	Processes instructions and controls the computer.	AMD Ryzen
3	RAM	Temporarily stores data being used by the CPU.	Corsair RAM
4	GPU	Processes graphics, video, and images.	MSI GeForce RTX
5	Storage (SSD)	Permanently stores the OS, software, and files.	Samsung 970 EVO Plus
6	Cooling System	Removes heat and prevents overheating.	AMD CPU cooler

➤ Conclusion: All six components work together to provide processing, memory, graphics, storage, connectivity, and cooling for the desktop PC.

Task 2: Motherboard Ports & Socket Identification

- Identify and label all major motherboard ports (SATA, power, USB headers, etc.).
- Identify all expansion slots (PCIe, M.2, etc.) and their purpose.
- Identify the CPU socket type and note which CPU families are compatible with it.
- Photograph or diagram the motherboard with labels.

Motherboard (Labeled Diagram)

Major Motherboard Ports & Connectors

24-pin ATX Power Connector Supplies main power to the motherboard.	8-pin CPU Power (EPS) Supplies power to the CPU.	SATA Ports Connects HDDs and 2.5" SSDs.	USB Headers Connects front-panel USB ports.
Front Panel Header Connects power/reset buttons and LEDs.	CPU Fan Header Connects CPU cooling fan.	System Fan Headers Connects case fans.	Audio Header Connects front-panel audio.
RGB/ARGB Headers Connects compatible lighting devices.	Rear USB Ports Connects external USB devices.	LAN/Ethernet Port Connects to a network (Internet/LAN).	Audio Ports Connects speakers, headphones and microphone.
HDMI/DisplayPort Connects a monitor (if supported by CPU).			

Expansion Slots

PCIe x16 Slot Graphics card (GPU) and other high-speed cards.	PCIe x1 Slot Wi-Fi, sound, capture and other expansion cards.	M.2 Slot High-speed NVMe SSD; some slots also support Wi-Fi modules.	DIMM Slots (RAM) Install RAM modules.
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CPU Socket

Socket Type: AM4
Compatible CPU Families:
AMD Ryzen 1000 – 5000 series
(Compatibility depends on motherboard chipset/BIOS).

Note: Always check the motherboard manual or manufacturer website for exact compatibility.

Summary Table

Component / Port	Function / Compatible With
24-pin ATX Power	Main power to motherboard
8-pin CPU Power (EPS)	Power to CPU
SATA Ports	HDDs / 2.5" SSDs
USB Headers	Front-panel USB ports
Front Panel Header	Power/Reset buttons, LEDs
CPU Fan Header	CPU cooling fan
System Fan Headers	Case fans
Audio Header	Front-panel audio
RGB/ARGB Headers	Lighting devices
Rear USB Ports	External USB devices
LAN/Ethernet Port	Network (Internet/LAN)
HDMI/DisplayPort	Monitor (if supported)
PCIe x16 Slot	GPU and high-speed cards
PCIe x1 Slot	Expansion cards
M.2 Slot	NVMe SSD / Wi-Fi modules
DIMM Slots	RAM modules
CPU Socket (AM4)	AMD Ryzen 1000–5000 series

➤ **Expansion Slots:**

Slot	Purpose
PCIe x16	Graphics card (GPU) and other high-speed cards
PCIe x1	Wi-Fi, sound, capture and other expansion cards
M.2 Slot	High-speed NVMe SSD; some slots can also support Wi-Fi modules
DIMM Slots	Install RAM modules

➤ **CPU Socket compatibility:**

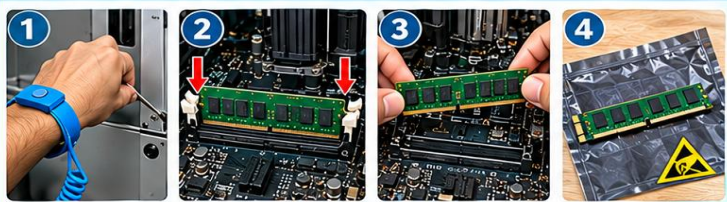
- **Socket:** AM4
- **Compatible CPUs:** AMD Ryzen 1000–5000 series (compatibility depends on motherboard chipset/BIOS).

➤ **Conclusion:** The motherboard provides power, connectivity, expansion slots, storage connections, and the CPU socket. The exact CPU compatibility and port layout depend on the motherboard's make and model.

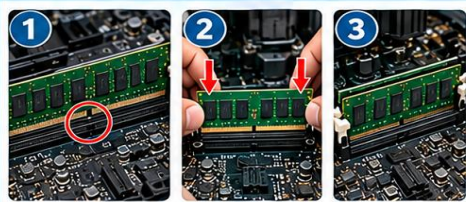
Task 3: RAM & Storage Installation

- Safely remove the RAM module, noting ESD precautions taken.
- Reinstall the RAM module correctly and confirm it is seated properly.
- Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).
- Confirm both components are detected correctly after reinstallation.

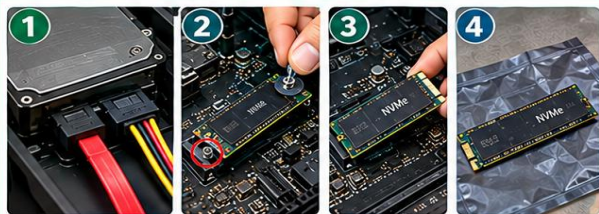
1. RAM Module Removal



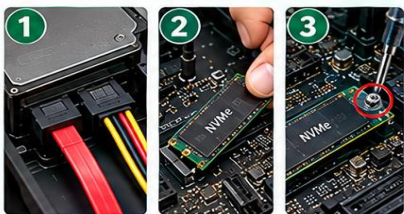
2. RAM Module Reinstallation



3. Storage Device Removal



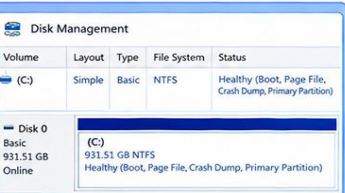
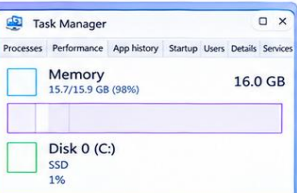
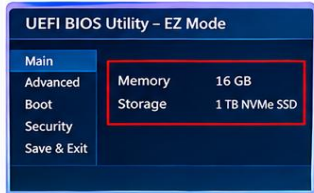
4. Storage Device Reinstallation



Connection Types Used



5. Verification



➤ RAM & Storage Installation:

1. RAM Removal and Reinstallation:

- Shut down the PC and disconnect the power cable.
- Wear an **anti-static wrist strap** or touch a grounded metal part to prevent ESD.
- Open the RAM slot clips and carefully remove the RAM module.
- Align the notch on the RAM with the slot.
- Press the RAM down firmly until the clips **click into place**.
- Check that both clips are locked and the RAM is seated evenly.

2. Storage Device Removal and Reinstallation:

- Disconnect the PC from power.
- For **SATA SSD/HDD**: disconnect the SATA data and power cables, remove the drive, then reconnect both cables securely.
- For **NVMe SSD**: remove the retaining screw, take out the SSD, insert it into the M.2 slot at an angle, and secure it with the screw.
- Ensure the connections are firm.

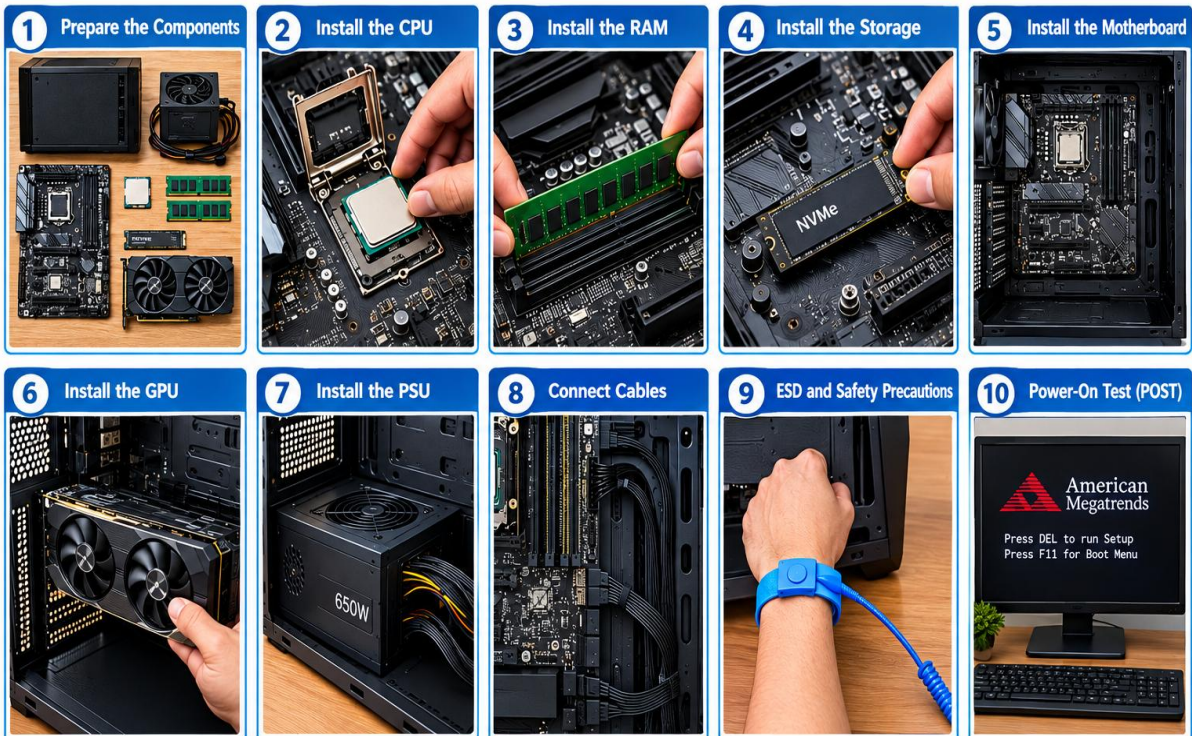
3. Detection Check:

- Power on the computer and enter **BIOS/UEFI**.
- Confirm the installed **RAM capacity** is displayed correctly.
- Check that the **SSD/HDD/NVMe drive** appears in the storage or boot-device list.
- Boot into the operating system and verify the storage drive is accessible.

➤ **Conclusion:** The RAM and storage device were safely removed and reinstalled using proper **ESD precautions**. Correct seating and BIOS/OS detection confirm that both components are installed properly.

Task 4: Full Desktop PC Assembly

- Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.
- Follow correct installation sequence and ESD/safety precautions throughout.
- Document each step with labeled photos or diagrams.
- Power on the system and confirm it passes POST successfully.



➤ **Full Desktop PC Assembly Steps:**

1. Collect all the PC components.
2. Install the CPU on the motherboard.
3. Insert the RAM into the memory slots.
4. Install the SSD or HDD.
5. Fix the motherboard inside the cabinet.
6. Install the GPU in the PCIe slot.
7. Install the PSU inside the cabinet.
8. Connect all power and data cables properly.
9. Follow ESD precautions to protect the components.
10. Turn on the PC and check that it passes POST successfully.

➤ **Conclusion:** The desktop PC was assembled successfully. All components were connected properly, and the system passed the POST test.

Section C — Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

1. identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.

➤ Hardware Components to Check First:

1. Power Supply (PSU)

- Check if the PSU is switched ON.
- Make sure the power cables are connected properly.
- A faulty PSU can stop the PC from starting.

2. RAM

- Check if the RAM is inserted correctly.
- Remove and reinsert it if necessary.
- Loose RAM can cause a boot failure.

3. CPU and Motherboard

- Check if the CPU is installed correctly.

- Make sure the CPU power cable is connected.
- Check the motherboard for loose connections or damage.

Conclusion: When a newly assembled PC does not boot, first check the PSU, RAM, and CPU/motherboard. These components are essential for the computer to start properly.

2. Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

➤ Motherboard–CPU Compatibility:

- Motherboard–CPU compatibility is important for the computer to work properly.

➤ For the system I assembled/inspected:

1. **Socket:** The CPU socket must match the motherboard socket.
2. **Chipset:** The motherboard chipset must support the CPU.
3. **BIOS:** The BIOS should support and recognize the CPU.
4. **Power:** The motherboard must provide enough power to the CPU.
5. **Performance:** Compatible components help the system run smoothly and avoid crashes.

If they are not compatible, the PC may not start, show no display, or work unstably.

- ### ➤ Conclusion: A compatible motherboard and CPU are necessary for the computer to boot correctly, work smoothly, and perform reliably.

3. Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.

➤ Verifying RAM and Storage Installation:

1. Check RAM

- Switch off the computer and unplug it.
- Make sure the RAM is properly fitted in the motherboard slot.
- Turn on the computer.
- Open **BIOS/UEFI**.
- Check that the correct **RAM size and speed** are detected.
- POST should start without memory error beeps or warning lights.

2. Check Storage

- Check that the **SSD/HDD cables** are properly connected.
- For an **NVMe SSD**, make sure it is properly inserted into the M.2 slot.
- Open **BIOS/UEFI** and check if the storage device is detected.

- After starting Windows, check **File Explorer or Disk Management** to confirm the drive is visible.

Conclusion: If BIOS detects the correct RAM and storage, and POST completes without errors, the components are installed correctly and ready to use.

4. Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.

➤ Hardware-Installation Checklist:

- ☐ **Check compatibility** – Make sure all hardware parts are compatible.
- ☐ **Install components properly** – Insert RAM, GPU, and storage devices correctly.
- ☐ **Connect power cables** – Check that all motherboard, CPU, GPU, and storage power cables are secure.
- ☐ **Check connections** – Make sure there are no loose cables or components.
- ☐ **Test the system** – Start the PC and check BIOS/POST for errors.

➤ Conclusion: Following this checklist helps prevent hardware installation mistakes and future PC failures.

5. Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

➤ SSD Instead of HDD:

➤ Why use SSD?

1. **Faster speed:** SSD starts the computer and opens applications quickly.
2. **Quick file access:** Office files, software, and documents load faster.
3. **Better reliability:** SSD has no moving parts, so it is less likely to be damaged by movement.
4. **Less noise:** SSD works silently.
5. **Lower power use:** SSD generally consumes less power than HDD.
6. **Better for office work:** It helps employees work smoothly with MS Office, web browsers, email, and other applications.

➤ **Conclusion:** For the lab's modern office workstations, **SSD is better than HDD** because it provides faster performance, reliability, and a smoother working experience.

Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI

- **Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.**
- **Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.**
- **Is presented as a clear checklist or simple decision flow a technician could follow on-site.**
- **Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue.**

➤ Step 1: Prompt given to AI:

- **Create a simple checklist for a junior technician to find why a newly assembled desktop PC will not boot. Include safety, PSU, motherboard, CPU, RAM, storage and POST. Put the steps in the correct order and keep the explanation simple.**

➤ **AI's checklist:**

1. Check power.
2. Check PSU cables.
3. Check motherboard.
4. Check CPU.
5. Check RAM.
6. Check GPU.
7. Check storage.
8. Turn on PC and check POST.
9. Check error codes.

➤ **My corrected checklist:**

1. Turn off the PC, unplug power and follow ESD safety.
2. Check the PSU and all power cables.
3. Check the motherboard and CPU installation.
4. Check CPU and motherboard compatibility.
5. Remove and reseat the RAM.
6. Check the GPU and its power cable.
7. Check the storage connection.
8. Turn on the PC and check POST/BIOS.
9. Check motherboard error lights or beep codes.
10. Test components one at a time if the problem continues.

➤ **What I Changed:**

- I added ESD safety as the first step because the AI checklist did not clearly mention it. I also added CPU and motherboard compatibility checking. These checks should be done before testing software because the problem occurred immediately after assembly. The corrected checklist is therefore safer and more suitable for a technician.